Phonon-Modulated Hubble Parameter H(t,Gamma) via F_U and E_net Coupling

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PAPER_1085: Phonon-Modulated Hubble Parameter $H(t,\Gamma)$ via F_U and E_{net} Coupling

Abstract

We derive the phonon-modulated Hubble parameter:

$$H(t,\Gamma) = H_0\left(1 + \frac{\Phi}{F_U}\cdot E_{\mathrm{net}}\right)$$

This couples the phonon resonance flux $\Phi_{1.25\,\mathrm{THz}}$ to the unified gravitational field F_U and the net vacuum energy E_{net}, producing a dynamical Hubble rate that amplifies during periods of strong phonon-vacuum coupling. Unlike PAPER_1084 (which derives the inflationary H from ρ_{SCm}), this formulation connects H to the present-epoch observables F_U, E_{net}, and Φ.

§1 Core Equation

$$H(t,\Gamma) = H_0\left(1 + \frac{\Phi(\Gamma)}{F_U(M,r)}\cdot E_{\mathrm{net}}(t)\right)$$

where: - $H_0 = 2.195\times10^{-18}$ s^{-1} (67.74 km/s/Mpc) - $\Phi(\Gamma) = \Phi_0\cdot S_{26}$ at resonance - $F_U = \sum_{i=1}^{26} GM/r^2\cdot[\mathrm{SSq}]\cdot i/26$ (26-layer gravity)

§2 Net Vacuum Energy

$$E_{\mathrm{net}}(t) = E_0\cdot e^{(\kappa+[\mathrm{SSq}]/26)t}\cdot S_{26}\cdot(2R-1)$$

where $R = F_{U,Bi}/F_U$ is the buoyancy ratio: - $R > 0.5$: expansion (positive E_{net}) - $R < 0.5$: erosion (negative E_{net})

§3 Amplification Factor

The phonon amplification of H is:

$$\frac{H(t,\Gamma)}{H_0} = 1 + \frac{\Phi\cdot E_{\mathrm{net}}}{F_U}$$

For solar parameters ($M = M_\odot$, $r = 1$ AU, $R = 0.8$): $H/H_0 \sim 10^6$, showing enormous amplification — physically this represents the local departure from Hubble flow near massive objects.

§4 Gamma Dependence

The phonon linewidth Γ enters through $\Phi(\Gamma)$:

$$\Phi(\Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\mathrm{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

At resonance: $\Phi = \Phi_0 \cdot S_{26} \approx 1.45 \times 10^{21}$. Off-resonance: Φ decreases Gaussian-sharply with linewidth.

§5 Physical Interpretation

The formula $H(t, \Gamma) = H_0(1 + \Phi/F_U \cdot E_{\mathrm{net}})$ encodes: 1. **Cosmic expansion base rate:** H_0 (Hubble constant) 2. **Local phonon enhancement:** Φ/F_U ratio (SCm vacuum coupling strength) 3. **Net energy direction:** E_{net} sign (expansion vs erosion) 4. **Linewidth modulation:** Γ controls phonon resonance sharpness

References

- PAPER_1084: SCm Phonon-Coupled Inflationary Scale Factor
- PAPER_1073: SCm Phonon-Driven Inflation
- PAPER_897: PhononModulatedEnergyEnetPhononCalc